

Decomposition of Concentration-Index Trees

Moses Mburu

1 Results

1.1 Exploratory Data Analysis

From the exploratory analysis, we see a high under-five mortality rate of 10.32% (Figure 1). This was not evenly distributed among subgroups. Among children who died before age five, 51.3% were from low-wealth households, compared with 42.8% among children alive at age five; 65.6% had no skilled birth attendance, compared with 56.8% among those alive; 69.3% were from rural areas, compared with 60.7% among those alive; and 67.5% had mothers with no education, compared with 55.2% among those alive (Table 3). The province distribution also shows that mortality was not evenly distributed across the DRC provinces, with the highest under-five mortality rate observed in Katanga (14.2%) and the lowest in Kinshasa (6.2%) (Figure 9).



Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

1.2 Decomposition of Concentration-Index

1.2.1 Indirect methods comparison (regression and predictive-forest SHAP)

From the indirect decomposition, we see that the largest regression contribution came from mother's education: 23.1% under CI and 28.7% under CIg/CIc (Table 1). Father's occupation in agriculture was second, contributing 17.9% under CI and 19.9% under CIg/CIc. In the predictive-forest SHAP decomposition, the contributions were less concentrated in one or two variables.

Residence and household wealth each contributed about 12% under CI, CIg and CIc, followed by mother’s occupation, father’s occupation, mother’s education and skilled birth attendance. The fitted regression model is reported in Table 4. Details of the fitted predictive forest are provided in Table 5 and Table 6, and its SHAP contributions are shown in Figure 10.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion

Variable	Regression			Predictive-forest SHAP			
	CI (%)	CIg (%)	CIc (%)	CI (%)	CIg (%)	CIc (%)	L (%)
Mother’s education	23.058	28.726	28.726	9.745	9.745	9.745	11.301
Father’s occupation: Agriculture	17.863	19.885	19.885	10.638	10.638	10.638	11.020
Mother’s occupation: Agriculture	11.684	14.393	14.393	11.333	11.333	11.333	11.892
Household wealth	9.116	8.788	8.788	11.955	11.955	11.955	7.723
Type of residence	8.733	11.866	11.866	11.958	11.958	11.958	12.522
Skilled birth attendance	7.848	9.988	9.988	8.627	8.627	8.627	7.620
Region: Equateur	3.761	1.066	1.066	4.842	4.842	4.842	4.393
Father’s occupation: Household, unskilled, not working	3.370	0.955	0.955	-0.901	-0.901	-0.901	-0.753
Region: Kinshasa	3.221	0.647	0.647	5.274	5.274	5.274	7.197
Region: Sud-Kivu	2.006	0.213	0.213	-0.397	-0.397	-0.397	0.121
Region: Kasai Oriental	1.746	0.513	0.513	0.582	0.582	0.582	0.763
Father’s education	1.462	0.869	0.869	1.948	1.948	1.948	2.271
Region: Maniema	1.075	0.083	0.083	2.223	2.223	2.223	1.820
Region: Orientale	0.984	0.251	0.251	0.026	0.026	0.026	-0.013
Region: Bas-Congo	0.678	0.060	0.060	0.040	0.040	0.040	0.117

1.2.2 Root-Node Inequality

Table 2: Root-node inequality in under-five mortality by concentration-index criterion

Root node	CI	CIg	CIc	L
Full sample	0.1156	0.0119	0.0477	0.1935

Tree validation plots

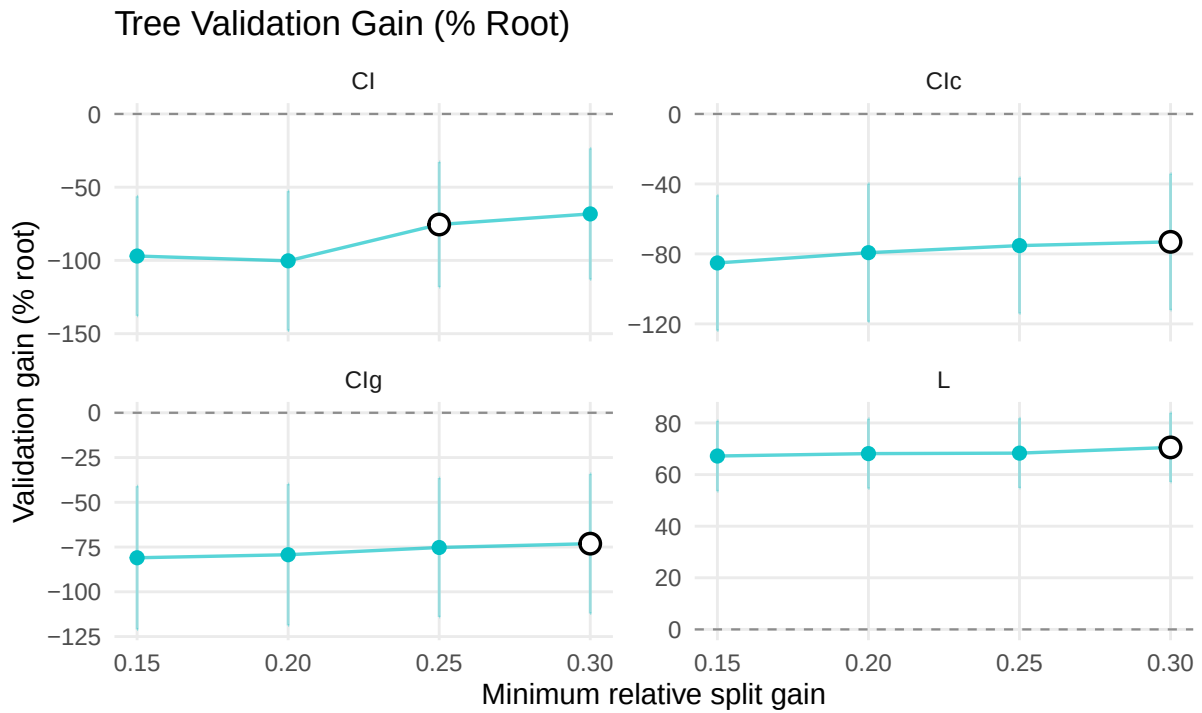


Figure 2: Tree validation gain and complexity by minimum relative split gain.

1.2.3 Selected Concentration-Index Tree

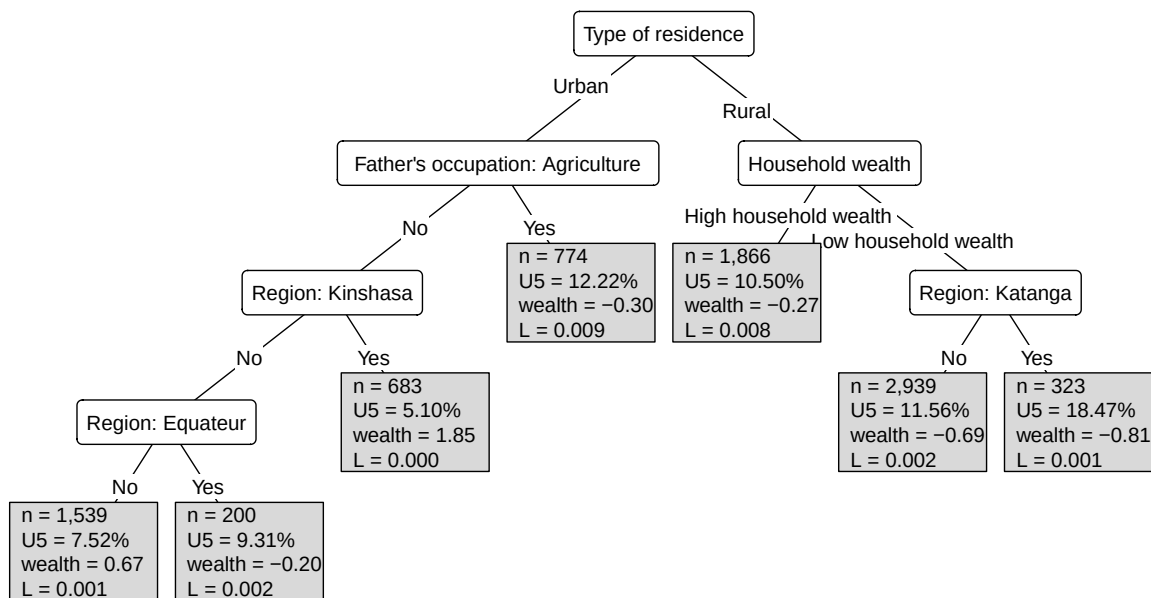


Figure 3: Selected concentration-index tree for under-five mortality in the DRC.

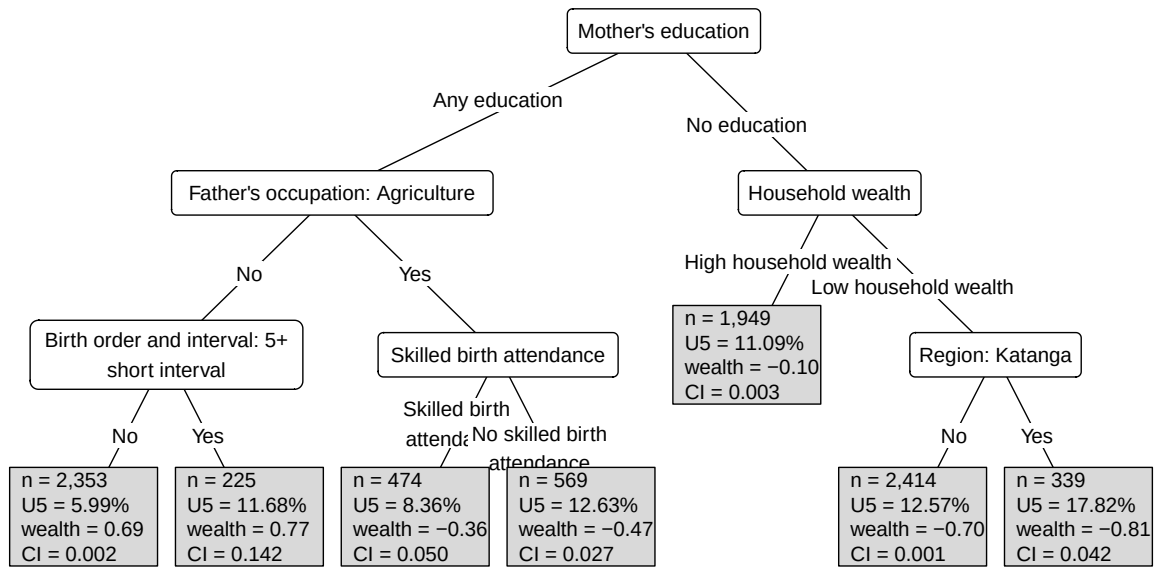


Figure 4: Concentration-index tree fitted with the CI criterion.

1.2.4 Surrogate Tree Comparison

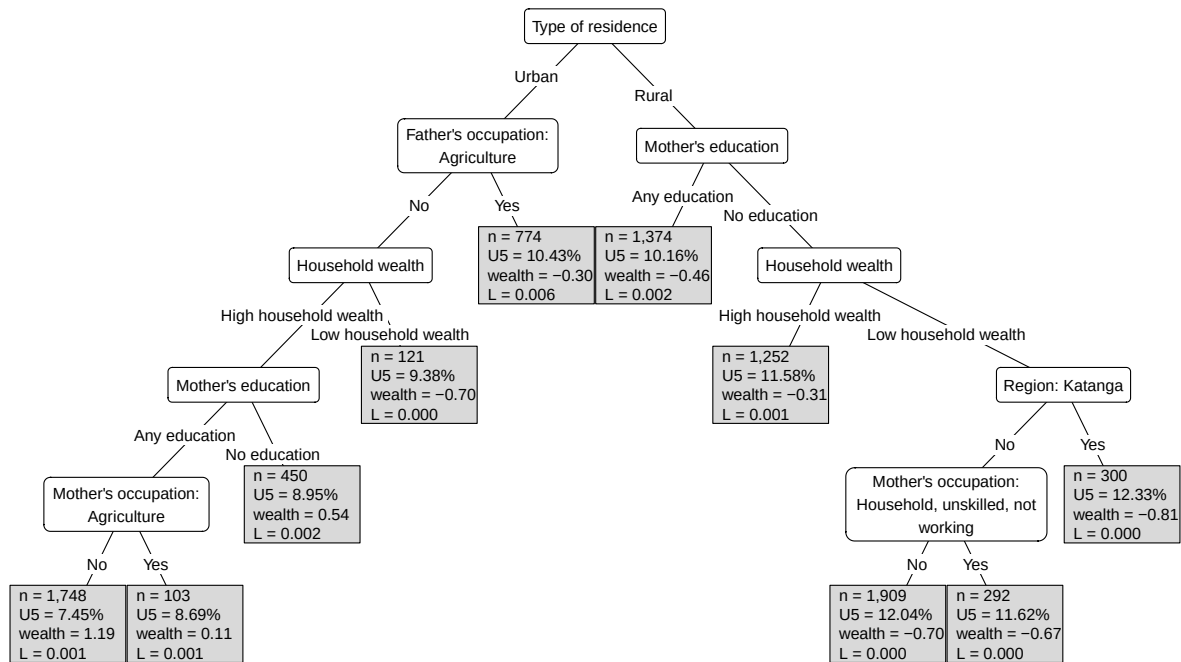


Figure 5: Surrogate tree summarising the selected concentration-index forest predictions.

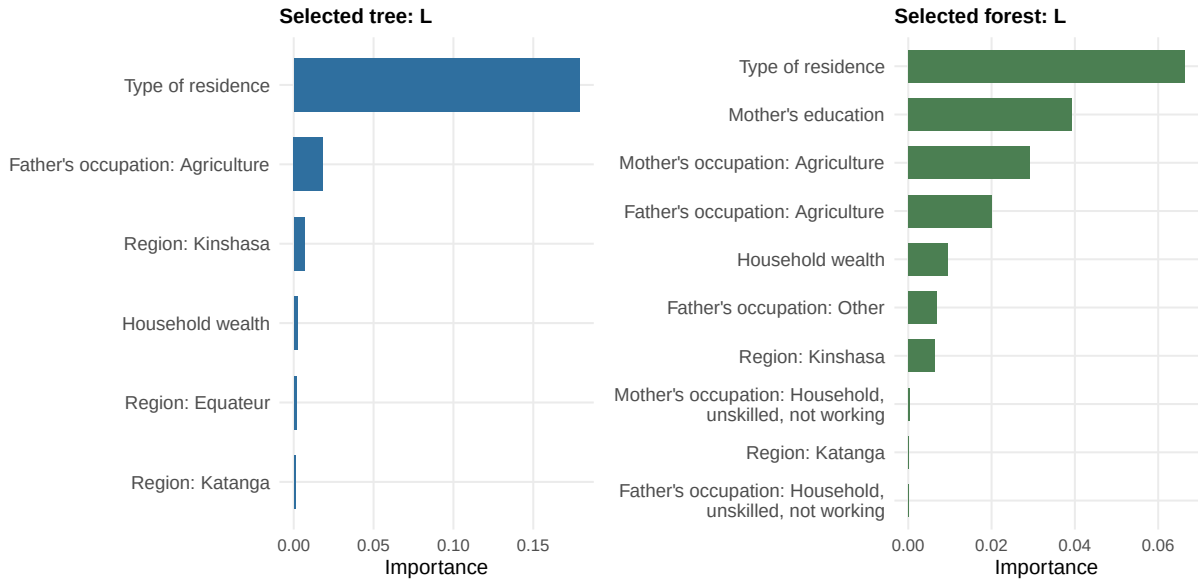


Figure 6: Variable importance for the selected concentration-index tree and forest.

1.3 Simulation Study

1.3.1 Recovery of the Simulated Subgroup Structure

The simulation study is used as a direct check on the tree-growing rule. In the null simulation, wealth, the binary outcome, and the candidate subgroup variables are generated independently. In the positive-control simulation, the outcome is concentrated among poorer observations in a planted subgroup defined by residence, maternal education, and province. The null trees should therefore remain at the root node, while the positive-control tree should recover a readable subgroup structure.

n = 3,000
U5 = 7.00%
wealth = 0.99
CI = 0.018

(a) CI

n = 3,000
U5 = 7.00%
wealth = 0.99
CIg = 0.001

(b) CIg

n = 3,000
U5 = 7.00%
wealth = 0.99
CIc = 0.005

(c) CIc

n = 3,000
U5 = 7.00%
wealth = 0.99
L = 0.000

(d) L

Figure 7: Null simulation concentration-index trees by criterion.

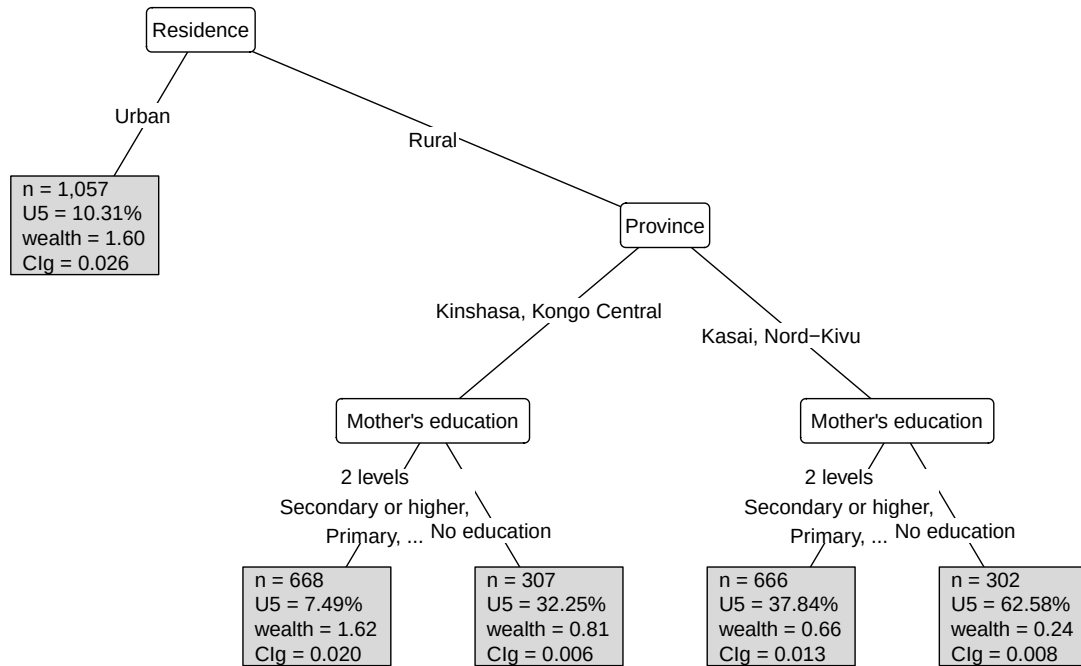


Figure 8: CIg concentration-index tree recovered from simulated data with a known subgroup-generating mechanism.

2 Appendix

2.1 Results

2.1.1 Appendix R1: Exploratory Data Outputs

Table 3: Weighted distribution of analysis variables by under-five mortality status

Variable	Under-five mortality		Overall
	Alive at age 5	Died before age 5	Overall
	(weighted N=7,464)	(weighted N=859)	(weighted N=8,323)
Household wealth			
High household wealth	4,266.9 (57.2%)	418.6 (48.7%)	4,685.5 (56.3%)
Low household wealth	3,197.4 (42.8%)	440.2 (51.3%)	3,637.6 (43.7%)
Skilled.birth.attendance			
Skilled.birth.attendance.1	3,225.3 (43.2%)	295.1 (34.4%)	3,520.5 (42.3%)
No.skilled.birth.attendance	4,239 (56.8%)	563.6 (65.6%)	4,802.6 (57.7%)
Sex of the child			

Female	3,828.8 (51.3%)	390.4 (45.5%)	4,219.2 (50.7%)
Male	3,635.5 (48.7%)	468.4 (54.5%)	4,103.9 (49.3%)
Birth order and interval			
First birth	1,396.8 (18.7%)	190.9 (22.2%)	1,587.7 (19.1%)
2-4 short interval	774.3 (10.4%)	110.1 (12.8%)	884.4 (10.6%)
2-4 long interval	2,603.8 (34.9%)	216.6 (25.2%)	2,820.3 (33.9%)
5+ short interval	696.6 (9.3%)	150.6 (17.5%)	847.2 (10.2%)
5+ long interval	1,992.8 (26.7%)	190.7 (22.2%)	2,183.5 (26.2%)
Mother's age at birth			
20 or more	4,256.1 (57.0%)	472.6 (55.0%)	4,728.8 (56.8%)
Less than 20	3,208.1 (43.0%)	386.2 (45.0%)	3,594.3 (43.2%)
Type of residence			
Urban	2,932.1 (39.3%)	263.6 (30.7%)	3,195.7 (38.4%)
Rural	4,532.2 (60.7%)	595.2 (69.3%)	5,127.4 (61.6%)
Mother's education			
Any education	3,342.9 (44.8%)	278.8 (32.5%)	3,621.8 (43.5%)
No education	4,121.4 (55.2%)	579.9 (67.5%)	4,701.3 (56.5%)
Father's education			
Any education	5,490 (73.6%)	589.1 (68.6%)	6,079.2 (73.0%)
No education	1,974.3 (26.4%)	269.7 (31.4%)	2,243.9 (27.0%)
Mother's occupation			
Other	1,610.2 (21.6%)	135.1 (15.7%)	1,745.4 (21.0%)
Household, unskilled, not working	1,757.2 (23.5%)	172.1 (20.0%)	1,929.4 (23.2%)
Agriculture	4,096.8 (54.9%)	551.6 (64.2%)	4,648.4 (55.8%)
Father's occupation			
Other	2,810.3 (37.6%)	242.1 (28.2%)	3,052.4 (36.7%)
Household, unskilled, not working	961.3 (12.9%)	108.5 (12.6%)	1,069.8 (12.9%)
Agriculture	3,692.7 (49.5%)	508.1 (59.2%)	4,200.9 (50.5%)

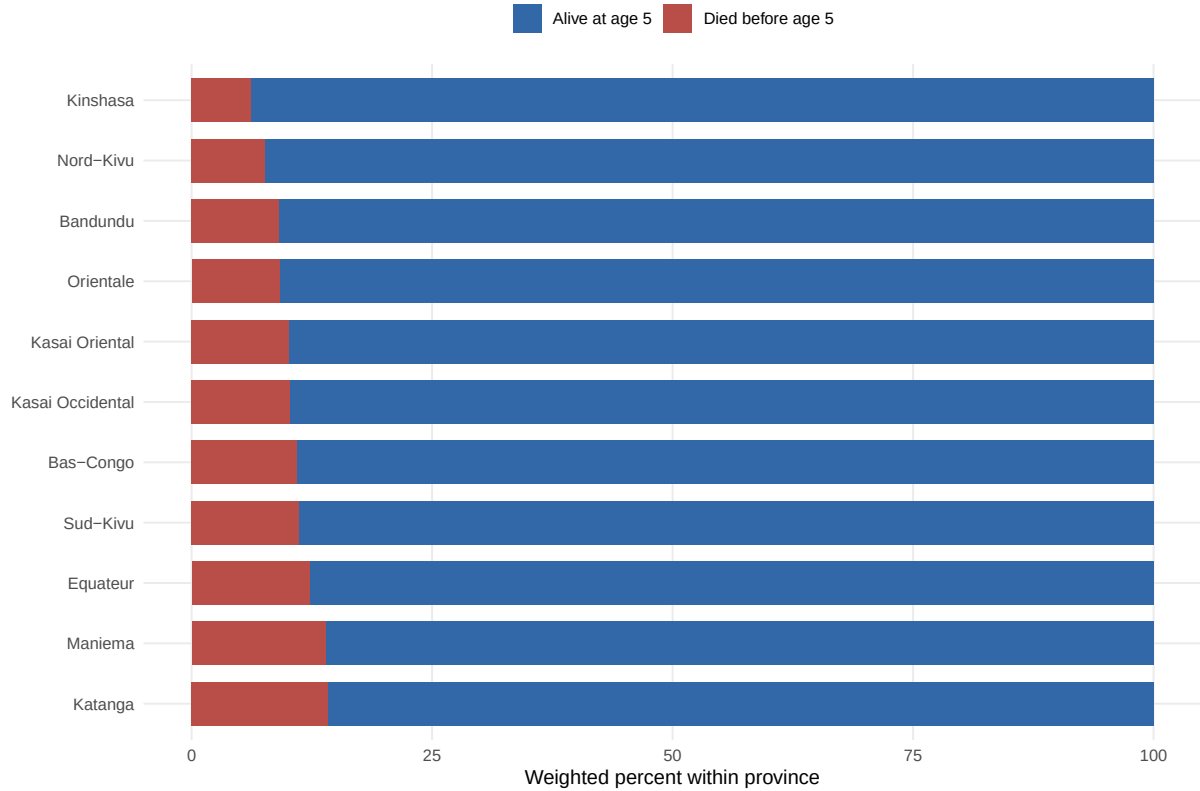


Figure 9: Weighted province distribution by under-five mortality status.

2.1.2 Appendix R2: Regression and Predictive-Forest Details

Regression model summary

Table 4: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.908	0.275	-10.563	<0.001	-3.448	-2.369
Household wealth	0.068	0.125	0.544	0.5865	-0.177	0.314
Skilled birth attendance	0.198	0.111	1.783	0.0746	-0.020	0.416
Sex of the child	0.242	0.105	2.294	0.0218	0.035	0.449
Birth order and interval: 2-4 short interval	-0.004	0.186	-0.020	0.9838	-0.367	0.360
Birth order and interval: 2-4 long interval	-0.509	0.169	-3.013	0.0026	-0.840	-0.178
Birth order and interval: 5+ short interval	0.420	0.210	1.995	0.0461	0.007	0.832
Birth order and interval: 5+ long interval	-0.396	0.211	-1.882	0.0599	-0.809	0.017
Mother's age at birth	0.033	0.143	0.229	0.8186	-0.247	0.312
Type of residence	0.090	0.154	0.584	0.5591	-0.212	0.392
Mother's education	0.336	0.132	2.555	0.0106	0.078	0.594

Father's education	-0.036	0.128	-0.280	0.7798	-0.286	0.214
Mother's occupation: Household, unskilled, not working	-0.018	0.155	-0.114	0.9093	-0.322	0.287
Mother's occupation: Agriculture	0.140	0.163	0.861	0.3895	-0.180	0.460
Father's occupation: Household, unskilled, not working	0.235	0.162	1.450	0.1472	-0.083	0.554
Father's occupation: Agriculture	0.225	0.149	1.509	0.1314	-0.067	0.517
Region: Bas-Congo	0.258	0.243	1.062	0.2883	-0.218	0.734
Region: Equateur	0.250	0.238	1.051	0.2934	-0.216	0.715
Region: Kasai Occidental	0.105	0.228	0.460	0.6452	-0.342	0.553
Region: Kasai Oriental	0.153	0.240	0.635	0.5251	-0.318	0.624
Region: Katanga	0.515	0.228	2.258	0.0240	0.068	0.962
Region: Kinshasa	0.081	0.262	0.310	0.7564	-0.433	0.596
Region: Maniema	0.489	0.224	2.178	0.0294	0.049	0.929
Region: Nord-Kivu	-0.252	0.294	-0.857	0.3915	-0.827	0.324
Region: Orientale	-0.084	0.265	-0.316	0.7522	-0.603	0.435
Region: Sud-Kivu	0.300	0.270	1.111	0.2667	-0.229	0.829

Predictive ranger tuning results

Table 5: Weighted class distribution before and after ranger undersampling

Sample	Outcome	Rows	Weighted N	Weighted percent
Before undersampling	Alive at age 5	7424	7464.30	89.68
Before undersampling	Died before age 5	872	858.80	10.32
After undersampling	Alive at age 5	1274	1289.74	60.03
After undersampling	Died before age 5	872	858.80	39.97

Table 6: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
accuracy	19	1100	0.6123	0.0037	10	1
accuracy	29	1100	0.6123	0.0042	10	2
accuracy	10	200	0.6090	0.0093	10	3
accuracy	29	200	0.6090	0.0106	10	4
accuracy	10	1100	0.6081	0.0024	10	5
accuracy	19	200	0.6034	0.0094	10	6
accuracy	1	200	0.5983	0.0014	10	7
accuracy	1	1100	0.5937	0.0003	10	8

accuracy	1	2000	0.5937	0.0003	10	9
accuracy	10	2000	0.5937	0.0003	10	10
accuracy	19	2000	0.5937	0.0003	10	11
accuracy	29	2000	0.5937	0.0003	10	12
roc_auc	1	200	0.6173	0.0118	10	1
roc_auc	10	200	0.6157	0.0110	10	2
roc_auc	1	1100	0.6147	0.0107	10	3
roc_auc	29	200	0.6126	0.0103	10	4
roc_auc	19	200	0.6119	0.0103	10	5
roc_auc	10	1100	0.6069	0.0105	10	6
roc_auc	29	1100	0.6053	0.0108	10	7
roc_auc	19	1100	0.6047	0.0108	10	8
roc_auc	1	2000	0.5000	0.0000	10	9
roc_auc	10	2000	0.5000	0.0000	10	10
roc_auc	19	2000	0.5000	0.0000	10	11
roc_auc	29	2000	0.5000	0.0000	10	12
sensitivity	29	200	0.3072	0.0162	10	1
sensitivity	19	200	0.2854	0.0168	10	2
sensitivity	10	200	0.2763	0.0159	10	3
sensitivity	29	1100	0.1100	0.0105	10	4
sensitivity	19	1100	0.0962	0.0134	10	5
sensitivity	10	1100	0.0608	0.0086	10	6
sensitivity	1	200	0.0126	0.0036	10	7
sensitivity	1	1100	0.0000	0.0000	10	8
sensitivity	1	2000	0.0000	0.0000	10	9
sensitivity	10	2000	0.0000	0.0000	10	10
sensitivity	19	2000	0.0000	0.0000	10	11
sensitivity	29	2000	0.0000	0.0000	10	12

Predictive ranger SHAP summary plot



Figure 10: SHAP summary plot for the predictive ranger model.

2.1.3 Appendix R3: Concentration-Index Tree Details

Tree tuning and selection details

Table 7: Best tree hyperparameters selected by validation gain within each impurity criterion

type	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	500	100	0.01	10	1e-04	0.25
CIg	500	100	0.01	10	1e-04	0.30
CIc	500	100	0.01	10	1e-04	0.30
L	500	100	0.01	10	1e-04	0.30

Tree fit and selection summaries

Table 8: Tree summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1191	NA
CI	Training gain	0.1013	0.0017
CI	Training gain (% root)	87.9025	1.5310
CI	Validation gain	-0.0235	0.0220
CI	Validation gain (% root)	-75.4763	42.7213
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0417	0.0008
CIc	Training gain (% root)	87.5995	1.4760
CIc	Validation gain	-0.0107	0.0134
CIc	Validation gain (% root)	-73.1067	39.0032
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0104	0.0002
CIg	Training gain (% root)	87.5995	1.4760
CIg	Validation gain	-0.0027	0.0034
CIg	Validation gain (% root)	-73.1067	39.0032
L	Root impurity	0.3128	NA
L	Training gain	0.1906	0.0061
L	Training gain (% root)	98.1536	0.2271
L	Validation gain	0.2861	0.1077
L	Validation gain (% root)	70.5254	13.4254

Tree selection metrics

Table 9: Selection metrics for fitted concentration-index trees

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Terminal nodes
CI	0.1191	0.1013	0.8790	-0.0235	-0.7548	7.5
CIc	0.0483	0.0417	0.8760	-0.0107	-0.7311	7.3
CIg	0.0121	0.0104	0.8760	-0.0027	-0.7311	7.3
L	0.3128	0.1906	0.9815	0.2861	0.7053	6.8

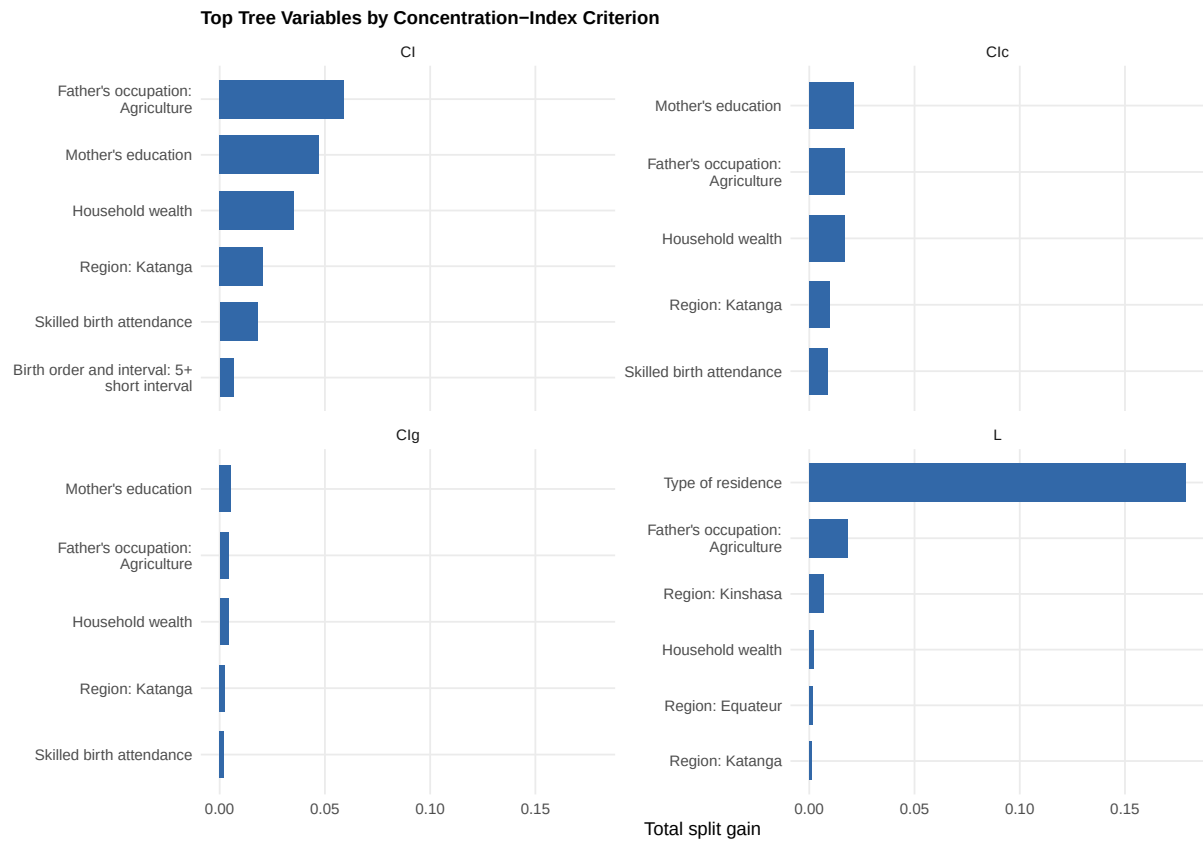


Figure 11: Top tree variables by concentration-index criterion.

Tree complexity plots

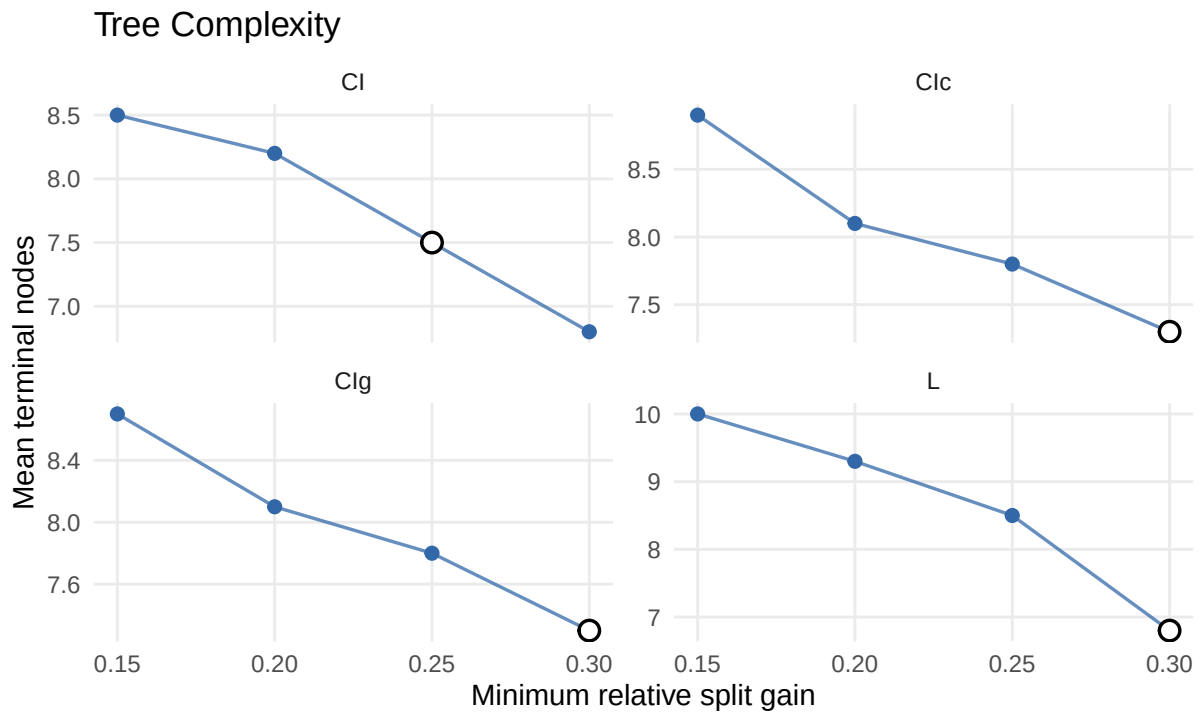


Figure 12: Tree complexity by number of terminal nodes.

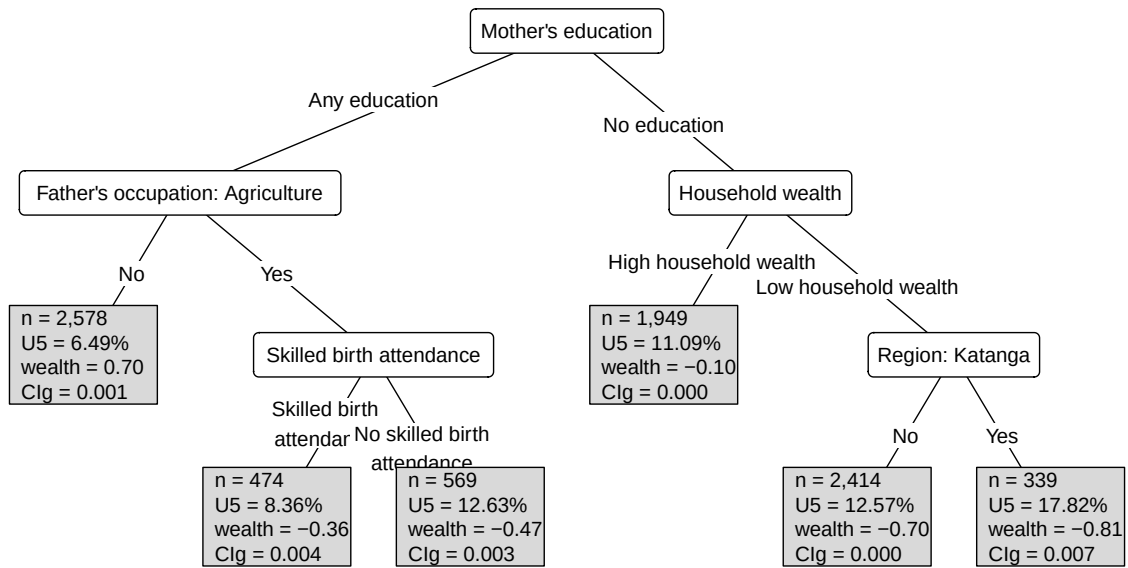


Figure 13: Concentration-index tree fitted with the CIg criterion.

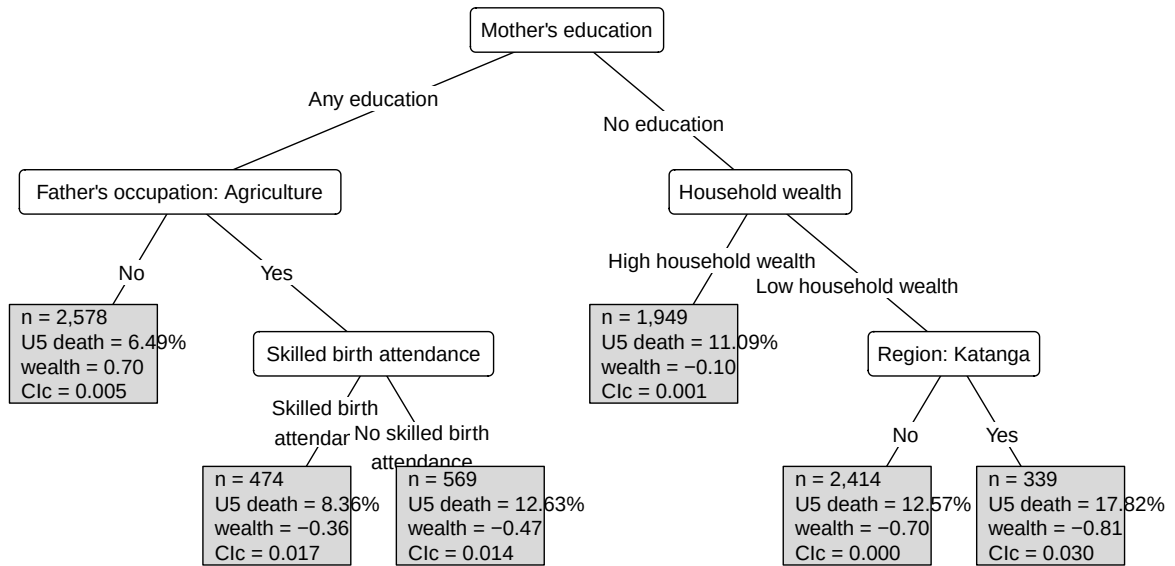


Figure 14: Concentration-index tree fitted with the Clc criterion.

2.1.4 Appendix R4: Rpart Comparison Tree Details

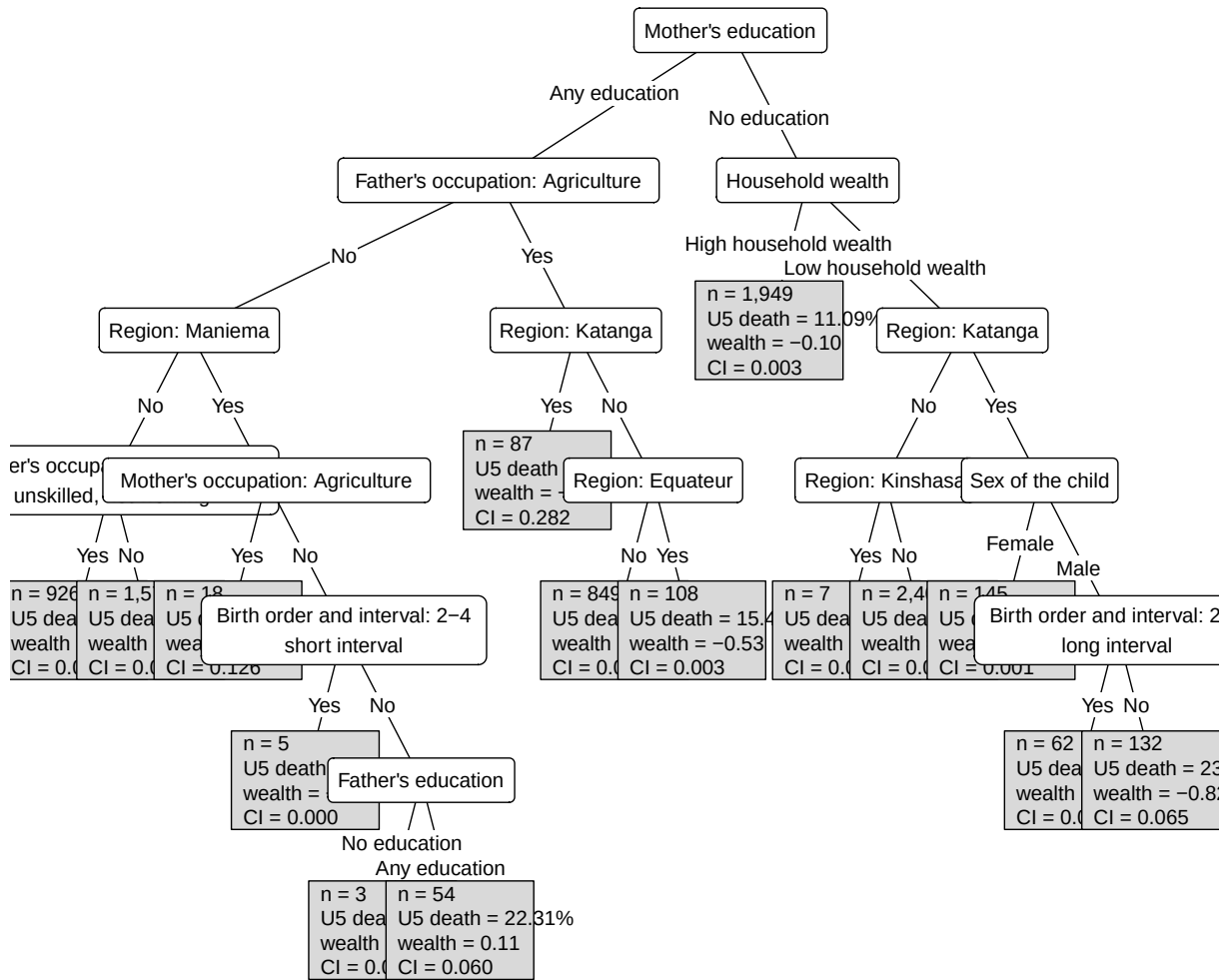


Figure 15: Lecturer rpart comparison tree using absolute CI impurity.

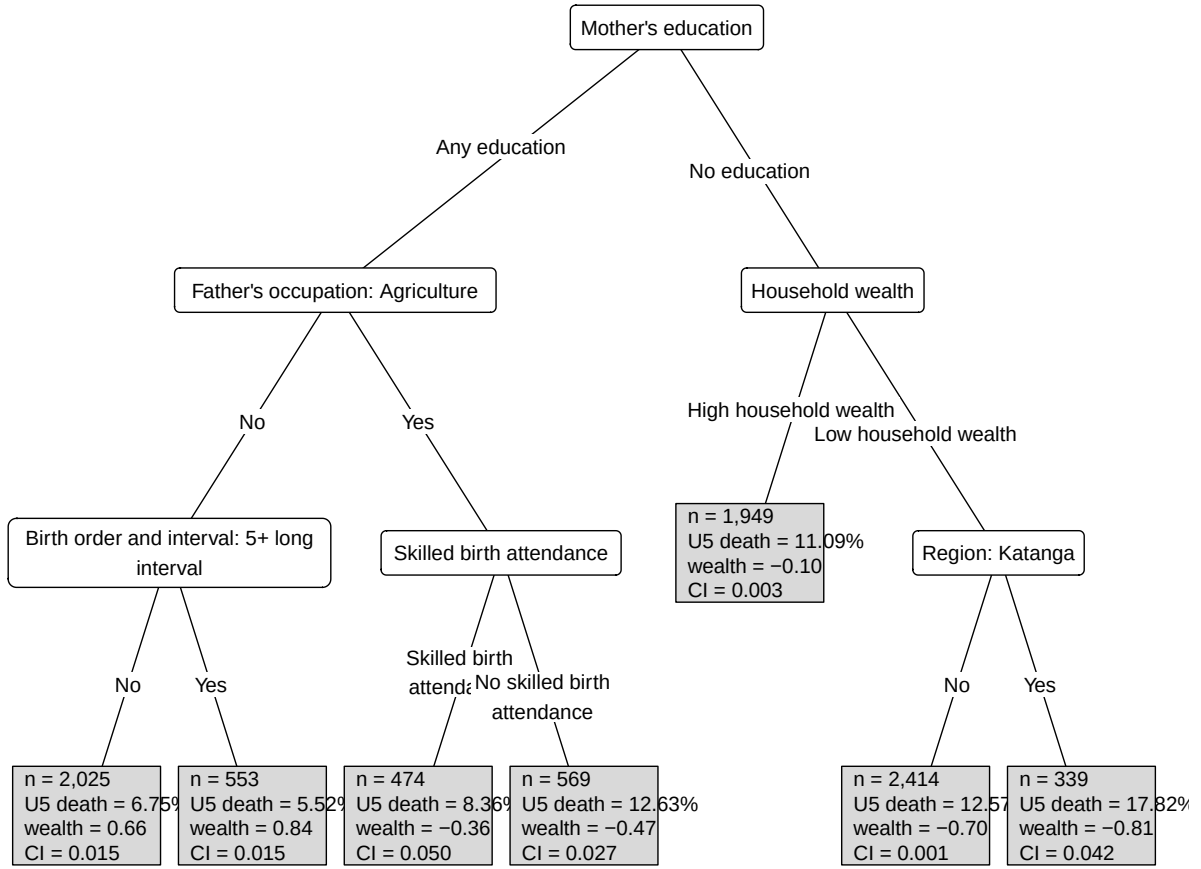


Figure 16: Lecturer rpart comparison tree using robust absolute CI impurity.

2.1.5 Appendix 5: Concentration-Index Forest Details

Forest tuning and selection details

Table 10: Best forest hyperparameters selected by validation gain within each impurity criterion

type	ntree	mtry	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	50	10	500	100	0.01	15	0	0.3
CIc	50	10	500	100	0.01	15	0	0.3
CIg	50	10	500	100	0.01	15	0	0.2
L	50	10	500	100	0.01	15	0	0.4

Forest fit and selection summaries

Table 11: Forest summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1191	NA
CI	Training gain	0.0222	0.0027
CI	Training gain (% root)	19.4745	2.4825
CI	Validation gain	-0.0032	0.0077
CI	Validation gain (% root)	-13.8147	8.1216
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0167	0.0007
CIc	Training gain (% root)	35.0907	1.5278
CIc	Validation gain	-0.0041	0.0052
CIc	Validation gain (% root)	-37.0209	19.7261
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0047	0.0002
CIg	Training gain (% root)	39.8170	1.2803
CIg	Validation gain	-0.0010	0.0014
CIg	Validation gain (% root)	-37.6958	20.8545
L	Root impurity	0.3128	NA
L	Training gain	0.1803	0.0059
L	Training gain (% root)	92.8313	0.4426
L	Validation gain	0.2803	0.1025
L	Validation gain (% root)	71.8405	13.3582

Forest selection metrics

Table 12: Selection metrics for fitted concentration-index forests

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Mean terminal nodes
CI	0.1191	0.0222	0.1947	-0.0032	-0.1381	1.846
CIc	0.0483	0.0167	0.3509	-0.0041	-0.3702	2.700
CIg	0.0121	0.0047	0.3982	-0.0010	-0.3770	3.196
L	0.3128	0.1803	0.9283	0.2803	0.7184	3.270

Forest validation plots

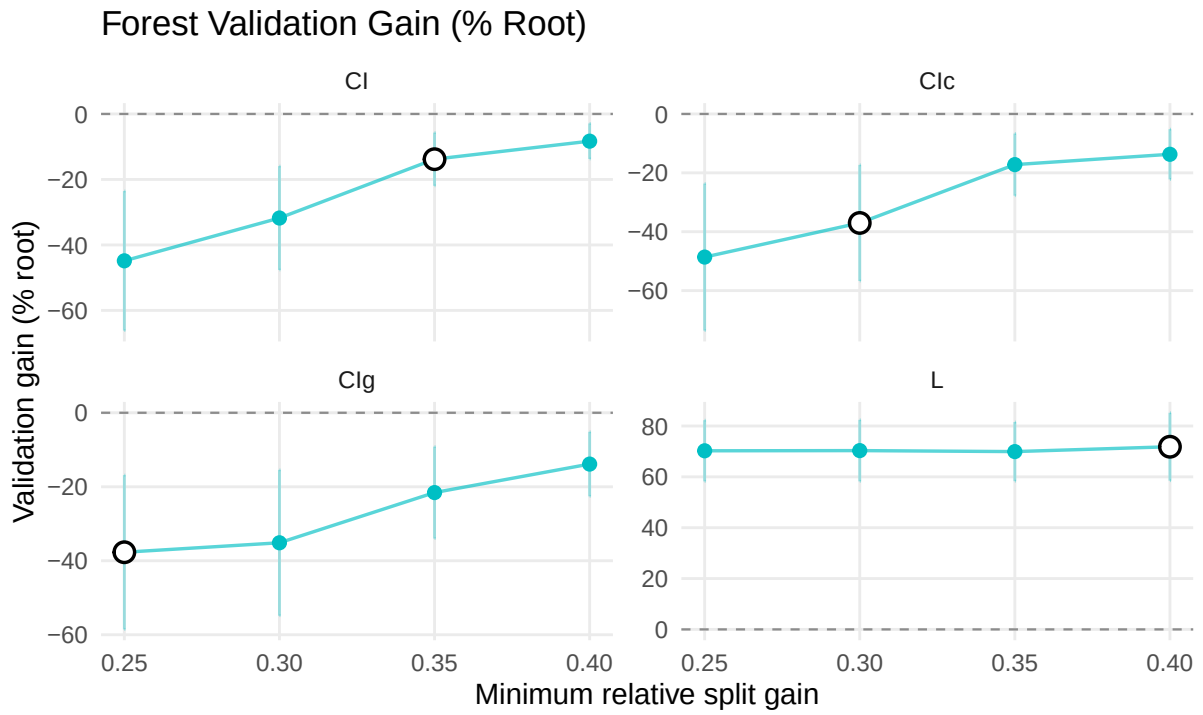


Figure 17: Forest validation gain and complexity by minimum relative split gain.

Forest complexity plots

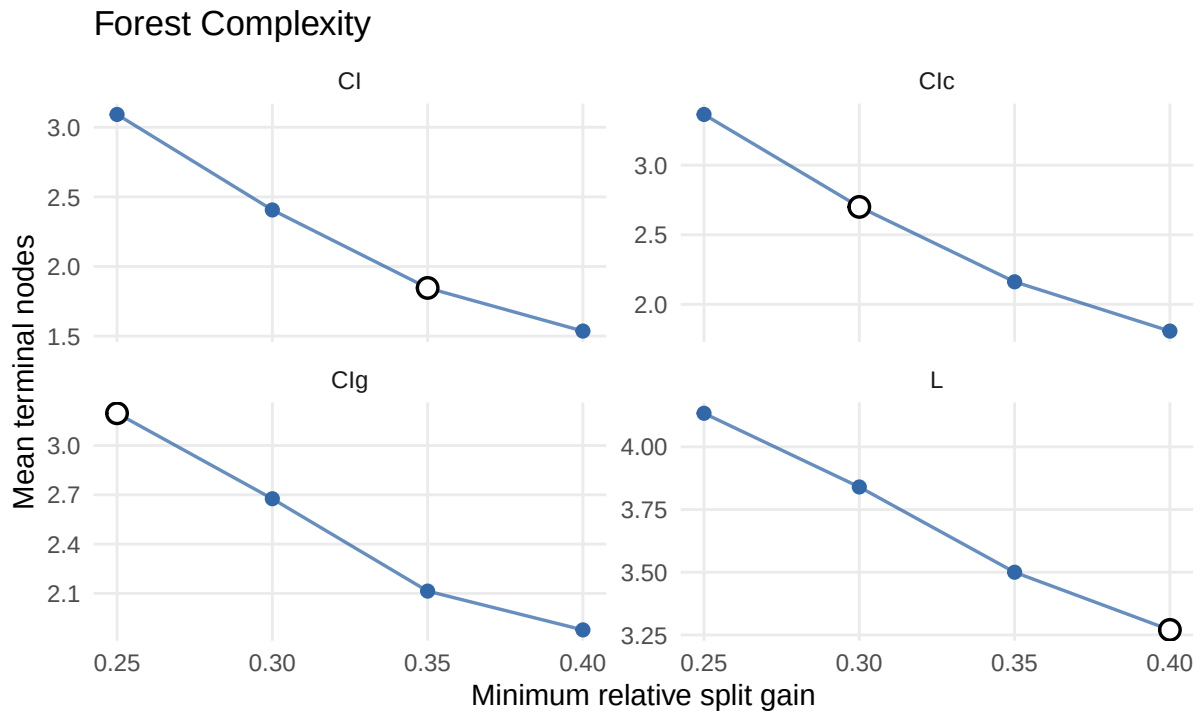


Figure 18: Forest complexity by number of terminal nodes.

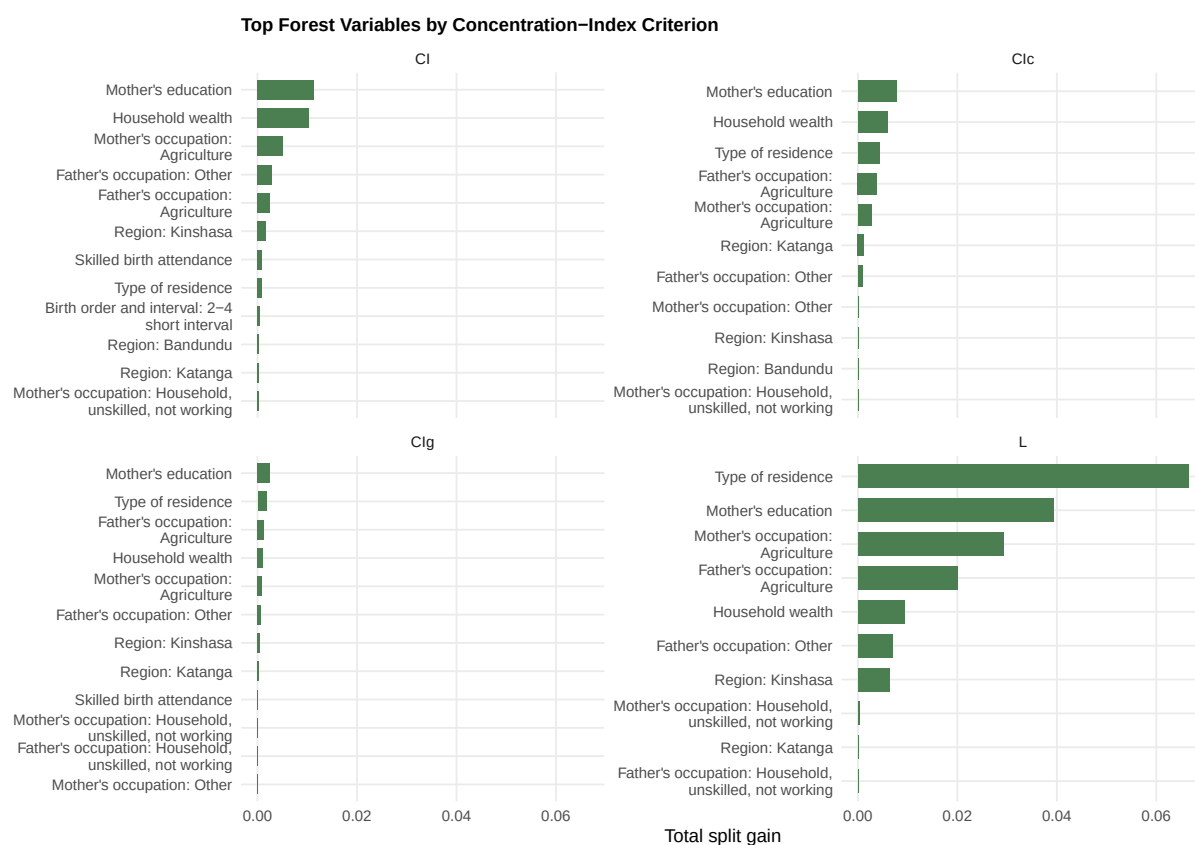


Figure 19: Top forest variables by concentration-index criterion.

Forest Surrogate Trees plots

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

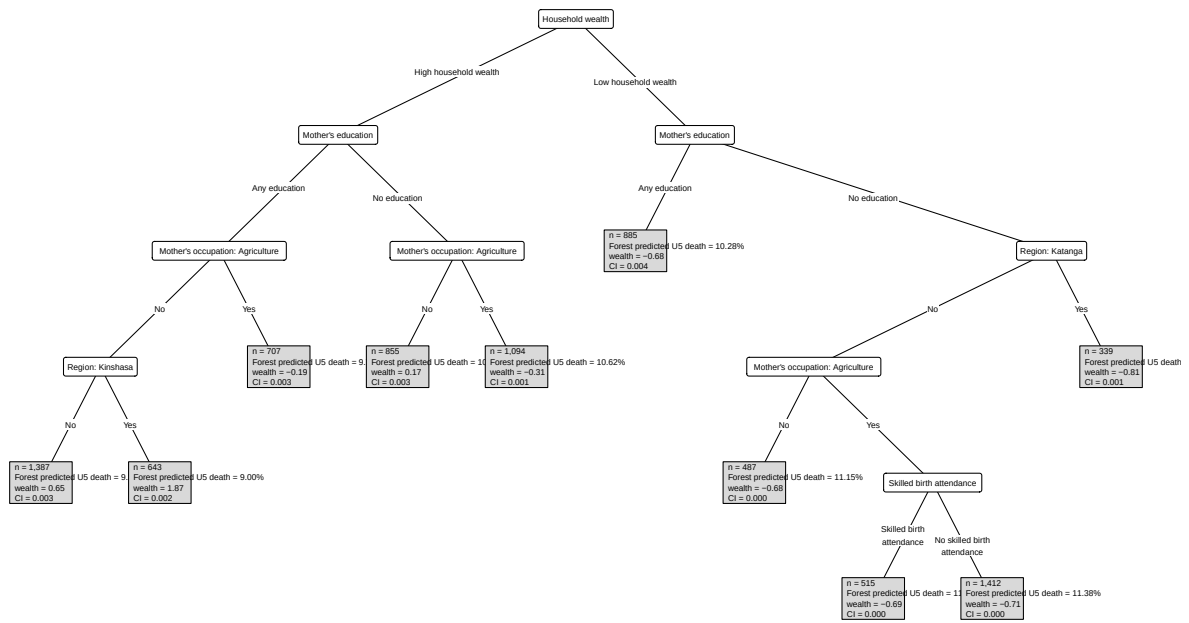


Figure 20: Surrogate tree for the CI concentration-index forest.

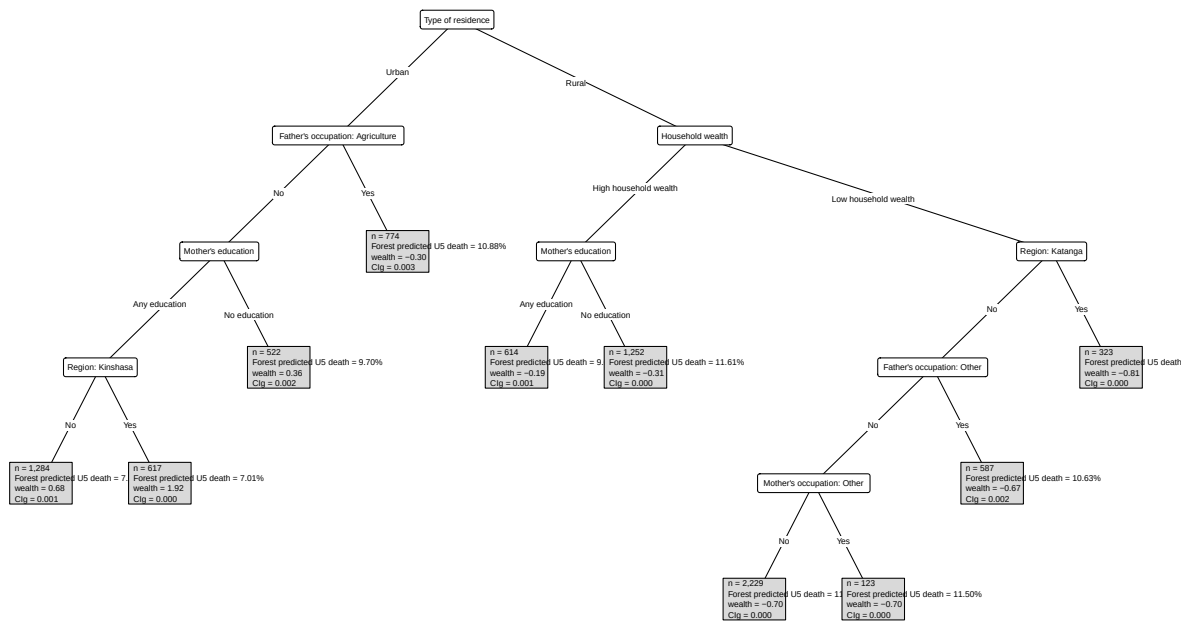


Figure 21: Surrogate tree for the Clg concentration-index forest.

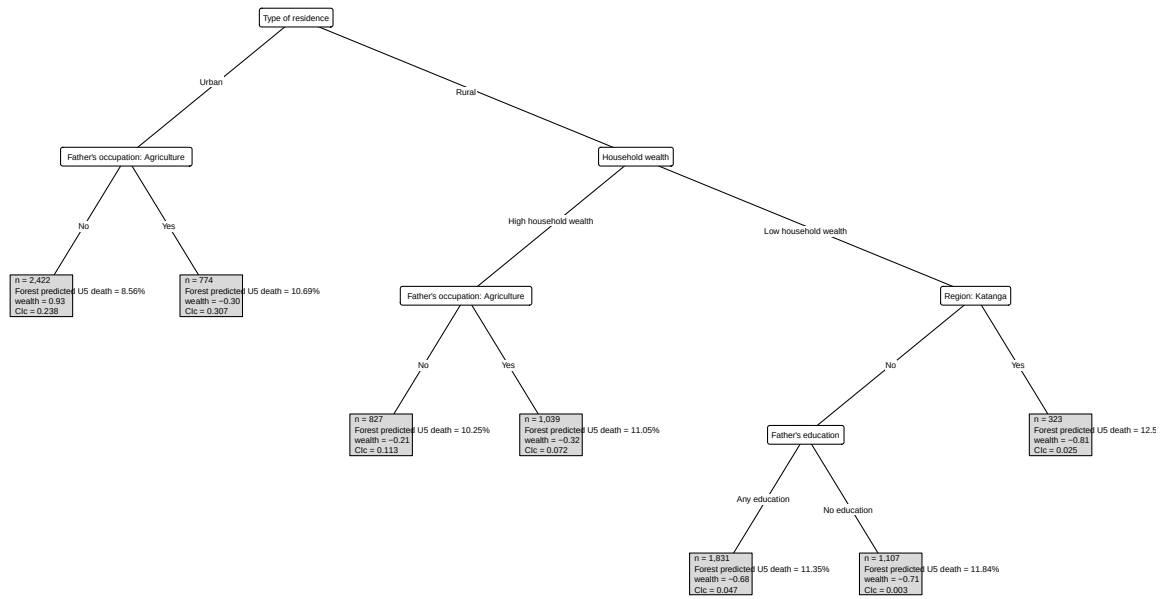


Figure 22: Surrogate tree for the CIc concentration-index forest.

2.1.6 Appendix S: Simulation Sensitivity Plots

The CIg tree is shown in the main results section. The remaining positive-control simulation trees are shown here to check whether the recovered subgroup pattern is stable across concentration-index criteria.

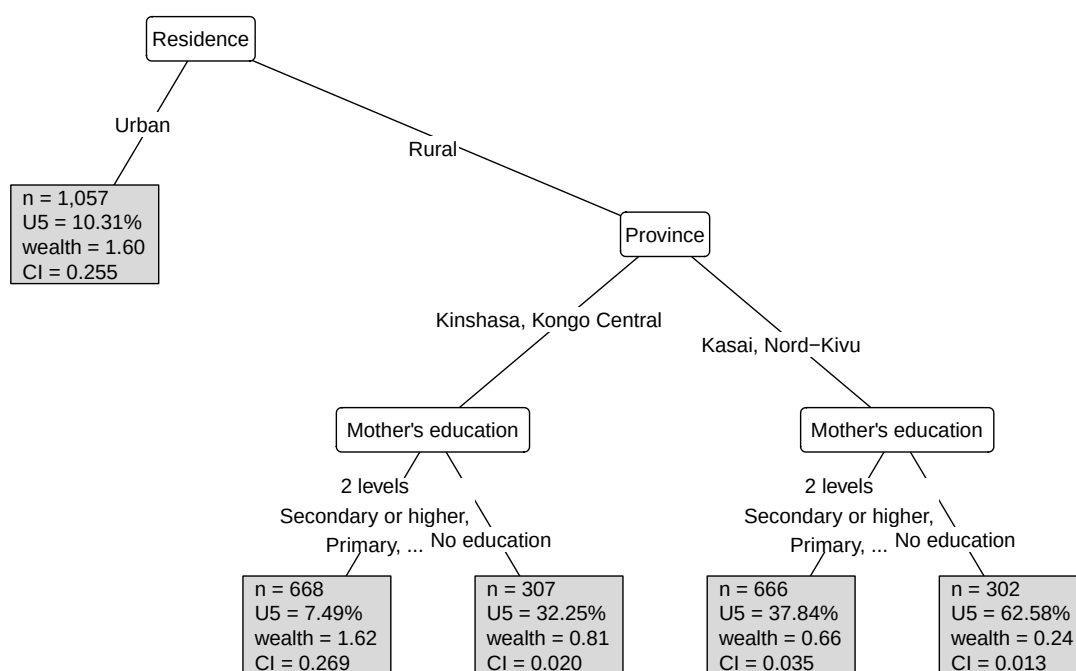


Figure 23: Positive-control simulation tree fitted with the CI criterion.

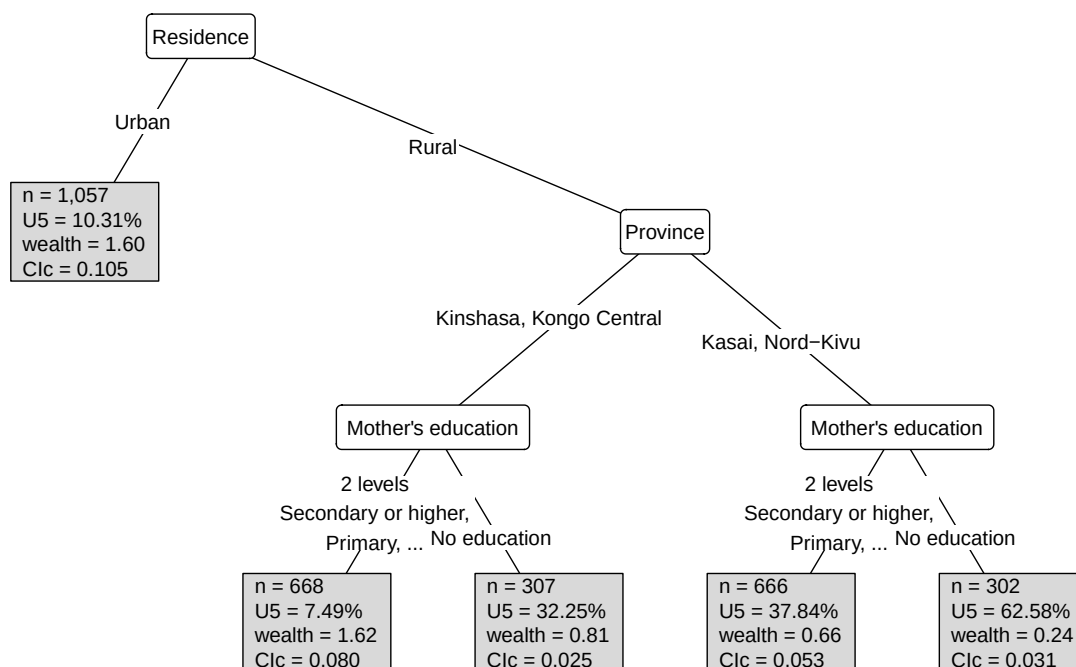


Figure 24: Positive-control simulation tree fitted with the Clc criterion.

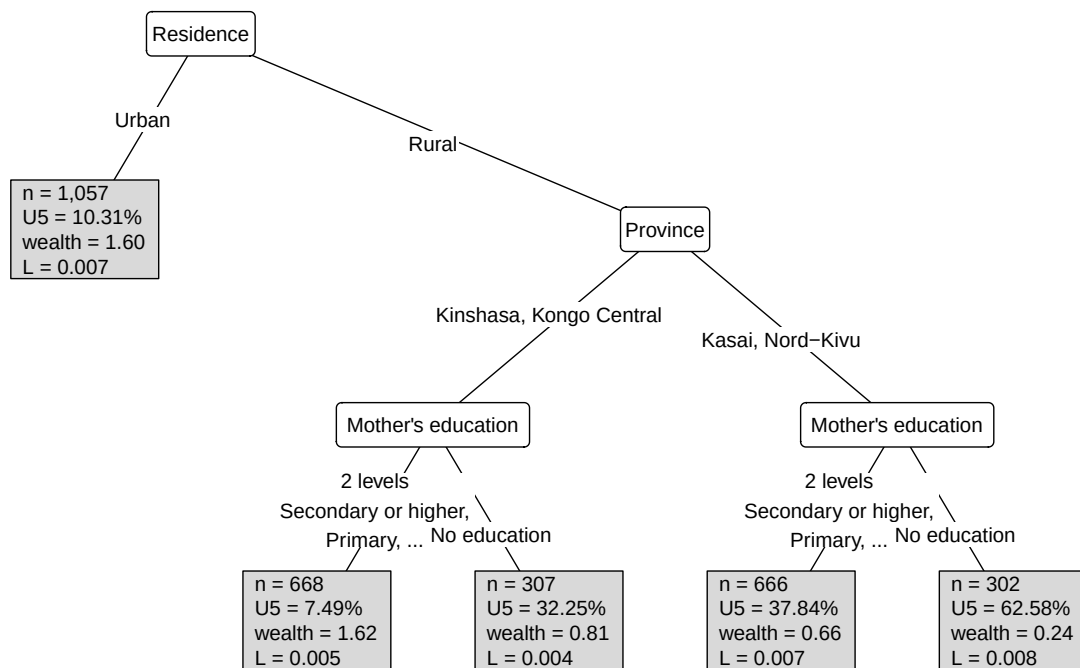


Figure 25: Positive-control simulation tree fitted with the L criterion.